BIOL 443: Bioinformatics Final Project

Part 1: Selecting a Project and Finding Sequencing Data

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Today's goal

You will select a biological system, choose a type of bioinformatics analysis, and locate publicly available sequence data for your final project.

By the end of class, you should have:

- A biological system or organism to investigate.
- One type of bioinformatics analysis.
- A dataset relevant to your project.
- A list of accession numbers identifying the data you plan to use.

Do not download any sequences or FASTQ files today.

Step 1. Choose your project type

Select ONE of the following:

- A. Genome comparison (DNA-seq): Compare DNA sequences or genetic variants among individuals, populations, or strains.
- B. Gene expression (RNA-seq): Compare which genes are expressed under different biological conditions.
- C. Microbiome analysis: Investigate bacterial or other microbial communities associated with organisms or environments.
- D. Phylogenetics: Compare DNA sequences to investigate evolutionary relationships among species, populations, or strains.

Step 2. Choose your organism or biological system

Choose an organism that interests you.

Examples:

- Human (*Homo sapiens*)
- Fruit fly (*Drosophila melanogaster*)

morganella morgaii

- Soybean (*Glycine max*)
- Honey bee (*Apis mellifera*)
- Mouse (*Mus musculus*)
- Corn (*Zea mays*)
- Bacteria or other microorganisms
- Another organism: Humans (Homo sapiens)

For microbiome projects, you may choose a host organism or an environmental sample (e.g., soil or water).

Scientific name of your organism (if applicable):

Step 3. Choose what you will compare

Complete ONLY the section corresponding to your selected project type.

A. Genome comparison

Select ONE:

- Two populations of the same species
- Two strains or varieties
- Two or more individuals

B. Gene expression (RNA-seq)

Select ONE:

- Control vs. stress treatment
- Healthy vs. infected
- Tissue A vs. tissue B
- Developmental stage A vs. stage B

C. Microbiome analysis

Select ONE:

- Diet A vs. diet B
- Healthy vs. diseased hosts
- Environment A vs. environment B

- Treatment vs. control

D. Phylogenetics

Select ONE:

- Evolutionary relationships among related species
- Genetic relationships among populations
- Relationships among bacterial or viral strains
- Evolutionary relationships among members of a gene family

Identify your groups or organisms:

Group/organism 1: Depressed human girl

Group/organism 2: Healthy human girl

For phylogenetics, you may include more than two species or groups.

Step 4. Find a sequencing study or sequence dataset

For genome comparison, RNA-seq, and microbiome projects

1. Open the NCBI Sequence Read Archive (SRA): https://www.ncbi.nlm.nih.gov/sra
2. Search for your organism's scientific name.
3. Add a keyword related to your comparison.
4. Open a relevant study.
5. Examine the study description and sequencing runs.

For phylogenetics projects

1. Open NCBI Nucleotide: https://www.ncbi.nlm.nih.gov/nuccore
2. Search for your organism and a gene of interest (e.g., COI, 16S rRNA, ITS).
3. Find sequences from related species or populations.
4. Make sure the sequences represent the same gene or homologous region.

Example searches

Project type	Example search
Genome comparison	Drosophila melanogaster genome sequencing
Gene expression	Glycine max drought RNA-seq
Microbiome	Apis mellifera gut microbiome 16S
Phylogenetics	Bombus COI

Search terms you used:

Step 5. Examine the dataset

A. For genome comparison, RNA-seq, and microbiome projects

Study title: Gut microbiome + Inflammatory Signatures in South african adolescent girls w/ depressive Symptoms

SRA study accession (SRP, ERP, or DRP): SRP

What type of sequencing was performed?

- Whole-genome DNA sequencing
- RNA-seq
- (•) 16S rRNA amplicon sequencing
- Shotgun metagenomic sequencing
- Other / Not sure

Which sequencing platform was used?

- (•) Illumina
- Oxford Nanopore
- PacBio
- Other / Not sure

Are the reads:

- (•) Paired-end
- Single-end

- Not sure

Does the study contain samples from the groups you want to compare?

- Yes
- No
- Not sure

Are there at least two independent biological samples per group?

- Yes
- No
- Not sure

B. For phylogenetics projects

Gene or genomic region selected:

- COI / COX1
- 16S rRNA
- ITS
- Another gene or whole genome: _____

Do the sequences represent the same gene or homologous region?

- Yes
- No
- Not sure

Do you have sequences from at least five organisms, populations, or strains?

- Yes
- No
- Not sure

Step 6. Record your accession numbers

Complete ONLY the table corresponding to your project type.

A. Genome comparison, RNA-seq, and microbiome

Find at least **four sequencing runs**, preferably representing two independent biological samples from each group.

Run accession numbers typically begin with **SRR, ERR, or DRR**.

Group	Sample description	Run accession number
Group 1	Depressed girl	SRR38158767
Group 1	Depressed girl	SRR38158768
Group 2	x depressed	SRR38158776
Group 2	x depressed	SRR38158783

Do these runs represent four independent biological samples?

- (•) Yes
- No
- Not sure

Note: Multiple sequencing runs may belong to the same biological sample.

B. Phylogenetics

Find at least **five DNA sequences** representing the same gene or homologous region.

Record their GenBank accession numbers.

Species / population / strain	Gene	GenBank accession number
1.		
2.		
3.		
4.		
5.		

Are all sequences from the same gene or homologous region?

- Yes

- No
- Not sure

Step 7. Final checklist

Before submitting, confirm:

- I selected a project type. ✓
- I selected an organism or biological system. ✓
- I identified what I want to compare. ✓
- I found a relevant dataset. ✓
- I recorded the study accession number (if applicable). ✓
- I recorded at least four SRA run accessions OR five GenBank sequence accessions. ✓
- I checked that the data are appropriate for my selected analysis. ✓
- I have not downloaded any FASTQ files or sequences yet. ✓

Submit this completed worksheet to canvas.

Your dataset must be approved before you begin downloading and analyzing sequencing data.